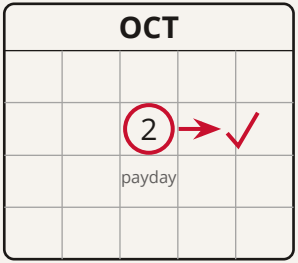
Kept

Promises customers can keep: one guarded action, due on payday, released by a person.



Team Sharpe Shooters
CIBC Collections Hackathon · 16 October 2026

THE PROBLEM

Collections today asks for money on the wrong day

- **Five systems, five keys per person.** Each files the same customer differently, so nobody sees the whole person.
- **Promises are dated by the calendar, not by pay.** A promise due before payday breaks; the bank's own data shows promised money paid is **+6.1 pts** higher when the due date falls within two days of payday.
- **More contact doesn't help.** The bank's randomised champion, two challengers and a hold-out keep promises at the same rate.
- **Hardship gets missed** in notes and calls, so vulnerable customers are pushed like anyone.

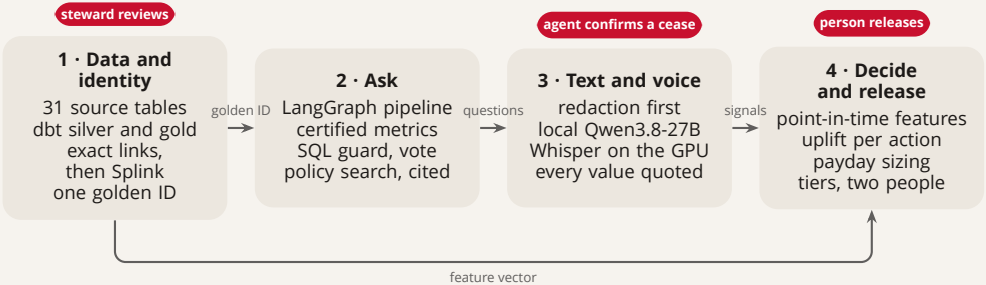
96,000
open cases tonight

$68.7M
balance at risk

38,115
need a hardship specialist

SYSTEM DESIGN

Four layers, one guarded decision



Nightly: dbt build → features → score 96,000 cards (~85 s). Live: the API rechecks every release on fresh data.

Source: src/kept/cli.py (nightly), docs/design/SharpeShooters_SystemDesign.pdf

ARCHITECTURE

A small stack that maps onto a bank's

Piece	Here	At the bank
Warehouse	DuckDB file, dbt silver/gold, Feast online store; 20x–40x today's data on 4 GB, never failed	the bank's lakehouse; dbt models carry over
Audit	DuckDB, append-only, SHA-256 chain, `kept audit-verify`; audit page 7.9 s → 0.04 s	WORM storage, same chain
Models	LightGBM uplift + promise, MLflow, gate; uplift, top 30%: 87 vs 64 extra payers per 1,000 for a response model, gain unproven; promise AUC 0.697 vs 0.625	model risk management sign-off
Local language models	Qwen3.8-27B on llama.cpp, Whisper, bge-m3; SetFit reads 760,594 notes in 63 s, the LLM would take 29 days	an on-prem inference cluster
The Ask's model	the same local Qwen3.8-27B, no hosted API or keys; 86 of 93 right in the pipeline, 36 with one prompt	the bank's approved model
API and app	FastAPI; Next.js workspace; personas; portfolio overview ~40 ms	SSO, the bank's design system

Speed: 96,000 cards in ~85 s; portfolio overview ~40 ms; audit page 7.9 s → 0.04 s after moving filters into SQL. 1,322 tests; every change through CI.

Source: README.md, pyproject.toml, docs/reports/decisions.md

LAYER 1 · DATA AND IDENTITY

Five keys become one person, and a steward checks the doubtful ones

- **Exact links first** (shared account and customer numbers), **then probabilistic matching** (Splink, names, dates of birth, addresses) for the rest.
- **Uncertain pairs go to a person**: 36,279 wait in the steward queue, oldest first, before two records ever merge.
- **Data quality is a contract**: 31 tables checked on load (types, ranges, duplicates), issues reported, not hidden.

99%
of 200 injected duplicates found

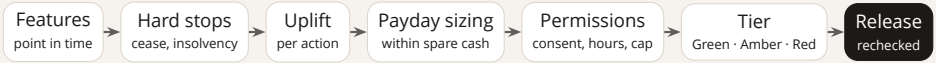
0
wrong merges

36,279
pairs for a steward, paged

Source: docs/reports/identity_recall.md, QUALITY_REPORT.pdf

LAYER 4 · DECIDE AND RELEASE

From a feature vector to a released action, in seven checked steps



Uplift, behind a gate

LightGBM T-learner: one model per action against waiting. Its signal is small (a call: +0.2 pts on a clean payment), so it only changes anything by passing the champion gate.

Due on payday

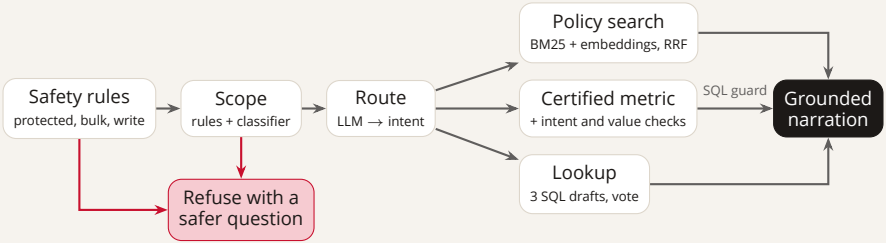
Next payday rolled forward from deposits or asked on the pay page; the promise falls due that day, the measured peak, inside spare cash.

People release

Green in a sampled team-lead batch; Amber by the agent; Red by a hardship specialist and a second person. Every release rechecks on fresh data.

LAYER 2 · THE ASK, AS A LANGGRAPH

Every answer walks a graph, and shows the path it took



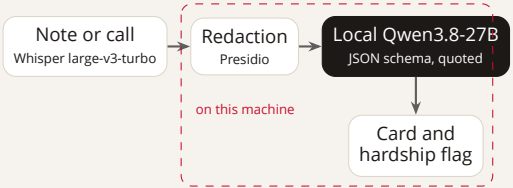
Every number in an answer must appear in the rows it came from; the SQL guard allows read-only queries on approved tables and columns only; the trace (*Route: metric · SQL guard: allowed · 1 row*) shows under each answer.

On the local model: released 21/21 · stress suite 38/42 · unseen drill 27/30. Two deterministic checks fix what the router can slip on: a channel the question names, a value the warehouse stores differently.

Source: src/kept/ask/graph.py, docs/reports/ask_accuracy.md

LAYER 3 · TEXT AND VOICE

Hardship heard the first time, without the words leaving the machine



- Every extracted value carries the exact quote it rests on; agents confirm a cease before it stops contact.
- Replies are decoded to the JSON schema: malformed replies went from 152 of 1,172 calls to 0.

1.00
hardship recall, notes and calls

9.4%
word errors (bank's speech-to-text: 12.4%)

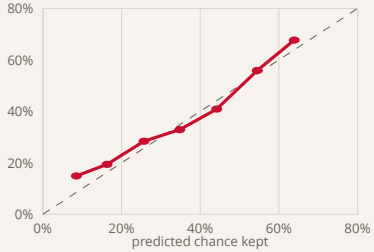
≥0.80
F1 on every extracted field

Source: docs/reports/extraction_gate.md, voice_wer.md

EXPLAINABILITY

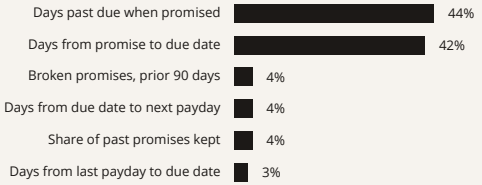
Every card says why, and the models say how far to trust them

Promise model: is a 60% promise kept 60% of the time?



AUC 0.611, Brier 0.220 on 50,653 promises it never saw (isotonic calibration). Dashed: perfect.

What drives it (share of |TreeSHAP|)



Per card: the 12 largest TreeSHAP contributions, both directions, on the case page. Uplift model: Qini 0.0008 ± 0.0003 on 141,350 held-out decisions.

Source: data/exports/reports/model_card.json (scripts/model_card.py)

DECISIONS, AND THE ALTERNATIVE WE REJECTED

Why it's built this way

We chose	Instead of	Because
Timing: payday and the call hour	more channels or intensity	the bank's A/B: strategies keep promises at the same rate; payday timing adds +6.1 pts; evening calls reach the right party +20.9 pts more
Uplift per action, behind a gate	a propensity-to-pay score	uplift asks the right question, but contact barely moves payment (a call +0.2 pts [-0.2 to +0.7]); the gate keeps a weak model out
Local models for everything	a hosted model	not cost (Haiku 5.5: USD 1,386 a year, ours 2,200); customer data stays inside (OSFI B-10, E-23, PIPEDA, Law 25)
Certified metrics first	free-form SQL from a model	router and certified metrics: 36 to 81 right of 93 (38.7% to 87.1%)
A graph with checks (LangGraph)	one prompt	86 of 93 right against 36; wrong but confident 33 → 7
Hardship by a rule; two people release Red	a model on the same cash-flow inputs	rule recall 0.652 vs model 0.672 at the same flag rate, no precision gain, and the model adds a gender flag
SetFit for note history	the LLM on every note	hardship F1 0.87 vs 0.88; 63 s for all notes vs about 29 days
Hybrid policy search (open)	dense only	dense MRR 0.747 vs 0.640 for hybrid (French 0.821 vs 0.564); hybrid still live
Schema-constrained decoding	parse-and-retry	malformed replies 152 → 0 of 1,172 calls
Qwen3.8-27B on llama.cpp	Gemma 3 12B on Ollama	Ollama ran a 4k window under a 9k prompt; released 18 → 21/21

RIGOUR

What we tried, measured, and didn't ship

A bigger local model

Gemma 4 26B on a 16 GB GPU: 24 s a note against 3 s, run-away replies, no better on held-out notes. Rejected; the dense Qwen3.8-27B fits and is faster.

A tuned uplift model

40 candidates tuned on a validation window; the best lost to the champion on held-out decisions (Qini 0.0029 vs 0.0073). X- and DR-learners won some arms and lost others. The gate kept the champion. Also: the bank's recommendation field is look-ahead, not a baseline; the gate's noise understated seed spread (call Qini +0.0023 to +0.0063).

A leaky feature

amount_ratio made AUC 0.92: it encoded the outcome. Dropped; honest AUC 0.63. A wider model then matched the bank's (0.697 vs 0.702); averaged with it, 0.764.

The rule: a change ships only if it beats what's live by more than the noise, on data it never saw.

Source: docs/reports/tuning.md, evidence.md §3, PR #169

SAFE BY CONSTRUCTION

Checked every night, not promised once

0 / 96,000
cards change when every protected attribute is scrambled and the book rescored

0.87–1.00
contact ratio across 11 protected attributes; a minimum-events rule

1 chain
every recommendation, release and override hash-chained; one edit breaks it

1 switch
reverts everyone to the bank's strategy; a worse model can't go live

- **Roles:** agents see their own cases; analysts see aggregates only (groups under 10 suppressed, no single case); names masked for every role but compliance.
- **Drift:** 8 features drift, 5 to watch, 13 stable (PSI vs training); 1 payroll features never appear in training rows. Drift on a model input pauses promotion until a retrain is checked.
- **Voice biometrics, accent and pitch are never inputs**; accent is used only to check Whisper works equally well (9.1–10.1% across regions).

Source: docs/reports/fairness_report.pdf, separated_review.md, drift_report.pdf; kept audit-verify

ONE CUSTOMER

Meet Jane: her last promise fell due before payday, and broke

- Pay lands in 3 days and is down 27% on her average.
- She answers calls after 18:00, not in the morning slot the strategy uses.
- Her notes say her shifts were cut: a hardship signal nobody acted on.

Kept's card for Jane

Call, Tue 18:00–20:00
$140 a month × 3, due on payday, inside ~$160 spare cash

Why: pay lands in 3 days (+9 pts) · pay down 27% (+6) · answers after 18:00 (+4)

Red hardship: a specialist and a second person release it

Source: the guided demo (case CS-2026-118204)

THE PRODUCT

One workspace for every role

Queue and case
ranked by expected recovery; a cited case brief from the local model; the full "why"; contacts, promises and month-ends from the warehouse

Portfolio
tonight's book, the A/B as a forest plot, cohorts, insights with their SQL, models, fairness and drift

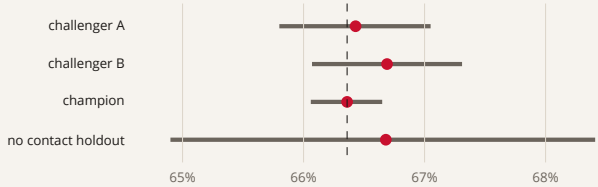
Ask, approvals, audit
plain-English questions with traced paths and sources; stage-by-stage release; the hash-chained trail

Notifications come from the audit log (nothing stored twice); the customer's pay page asks for payday when the bank doesn't know it (about 3 customers in 4).

MEASURED IMPACT

The bank's own randomised test: strategy doesn't move promises, timing does

Promise-kept rate by strategy (95% Wilson interval)



Every interval overlaps the champion's (dashed).

+6.1 pts

promised money paid when due 0–2 days after payday (95% CI 2.8–10.6), 12,808 promises in randomised cases

0 per 1,000

customers paying within 14 days gained by Kept choosing the action instead of what the bank did, clean outcome (95% CI -7 to +8): no measurable gain in who to contact

Source: silver.collections_cases (closed, randomised); evidence.md §1–2, §4

IS IT REAL?

Every claim, its test, its 95% interval

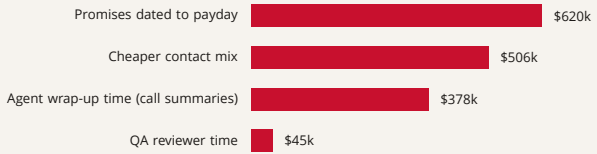
Claim	Test	Result [95% CI]	
Payday dating raises money paid	randomised cases	+6.1 pts [2.8–10.6]	holds
It is the payday, not the customer	vs 1–3 days either side	+7.7 [5.0–9.8] · +8.3 [6.1–10.5]	holds
Same customer, on vs off payday	fixed effects	+4.0 pts [1.1–7.4]	holds
Fake paydays show nothing	paydays moved 3–10 days	6 of 6 intervals include 0	holds
Promise model vs the bank's	paired bootstrap	0.697 vs 0.702 [-1.9 to +1.0]	tie
The two models together	rank average, nothing fitted	0.764, +6.1 [5.1–6.9]	holds
Kept's action vs what the bank did	doubly robust, clean outcome	0 payers / 1,000 [-7 to +8]	no gain
Evening calls reach the right party	comparable calls	+20.9 pts [+20.1 to +21.8]	holds
Targeting by uplift	randomised cells only	$+18K to $+42K / 1,000, CI spans 0	unproven
Pays back in year one	20,000 simulated draws	99.97% (98.5% if correlated)	holds

Source: evidence.md §2, §2a, §4, §8, §9; uplift_improvement.md; business_case.pdf

BUSINESS VALUE

$1.55M a year in the base case, and where it comes from

Base case by driver ($k a year)



Payday dating: more promised money paid on the promises made each year. Contact mix: fewer calls, more payment links. Wrap-up: call summaries agents don't type. QA: reviewers check flagged calls only.

$1.55M
a year, base case (low $0.44M, high $4.27M)

2.4×
year-one ROI, base case

2.6 months
payback, base case

99.97% of 20,000 draws over every assumption pay back in year one (98.5% if they all move together). With no payday effect it still pays back; with everything else at its low end it needs 0.2 pts.

Source: docs/reports/business_case.pdf, data/exports/reports/business_data.json

VALUE BEYOND DOLLARS

What the customer and the regulator get

Guard (must not get worse)	Bank today	Kept
Contact-cap breaches	0.03% of cases	0 (checked at release)
Contact outside calling hours	1.4% of cases	0
Hardship notes missed	not measured	recall 1.00
Any-contact fairness ratio	not measured	0.87 or above
Complaints	3.1% of cases	needs a live pilot cell

Every action is explained (fixed sentences, never model text), on the record, and released by a named person.

Source: evidence.md §5 (guard metrics)

NEXT

A pilot that proves it, with a rule that ends it either way

- **20% hold-out first**: the payday effect shows within 2 weeks (80% power); then a permanent 5% on the bank's strategy (7 weeks to the same answer).
- **A randomised due-date cell**, so the payday effect becomes a trial, not an estimate.
- **Primary metric:** promised money paid. **Guards:** complaints, caps, hours, fairness ratio.

Decision rule

Stop early for harm if any guard worsens beyond its 95% interval; scale up if promised money paid beats the hold-out's interval for two consecutive months; otherwise run to the planned size.

Day one

kill switch, nightly drift and fairness checks, the audit chain, and payday asked where the bank doesn't know it.

HOW WE BUILT IT

Four tracks, gated, every number reproducible

- Data · Ask · Text and GPU · Decisions, integrated at gates G0–G4.
- 170+ pull requests, each through CI; 1,322 tests.
- Local models only: no hosted API and no keys; nothing leaves the machine.
- `JUDGES.md`: every claim, its report, and the command that reproduces it.

Reproduce any number

```
uv run kept nightly
uv run python scripts/evidence_report.py
uv run python scripts/model_card.py
uv run python scripts/bench_run.py ...

bash scripts/secret_scan.sh
```

Jane’s promise falls due on her payday, inside what she can spare, and her hardship was heard the first time she said it.

APPENDIX

Who can see and do what

Role	Sees	Names	Releases	Ask
Collections agent	own cases	masked	Amber on own cases	yes
Hardship specialist	every case	masked	Red, first approval	yes
Team lead	every case	masked	Red second; Green batch	yes
Compliance	every case and the full trail	unmasked	Red second	yes
Analyst	aggregates only (groups ≥ 10)	none	none	yes
Data steward	match pairs	masked	merge or split	yes

Source: src/kept/governance/access.py, src/kept/web/src/lib/session.ts

APPENDIX

The Ask's benchmarks, on the local model

21/21
the organisers' released questions with dev gold

38/42
our stress suite, aimed at the release's traps

27/30
an unseen drill we never tuned on (one question in French)

- Every model call is local (Qwen3.8-27B on llama.cpp): no hosted API, no keys, no quota. The graph's deterministic checks (channel, stored values, column repair, dates, cohorts, relevance, "can we" means policy) and a convention linter keep the model on the rails; drafts that disagree escalate to five, then the model judges the results.
- A refusal must refuse; a number must match gold within the stated tolerance; a policy answer must cite the section.

Source: data/exports/reports/{released,stress,drill}_answers.json; docs/reports/ask_accuracy.md

APPENDIX

The uplift search we ran, and why the champion stayed

Model, on held-out decisions	Qini, calls	noise	Qini, any contact
Champion (live)	0.0073	0.0003	0.0008
Tuned challenger	0.0029	0.0003	-0.0006

40 candidates were compared on a validation window inside the training period (15 Jul to 15 Aug); only the best was judged once on the test decisions, by the nightly's own gate. It lost beyond noise, so the champion stays.

Source: docs/reports/tuning.md (scripts/tune_uplift.py)

APPENDIX

What the uplift study found

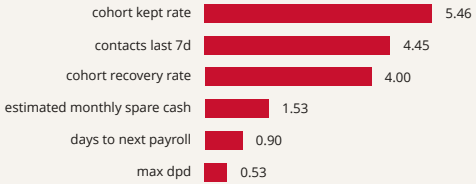
Finding	Number
Contact barely changes payment	a call adds +0.2 pts [-0.2 to +0.7] on a clean payment (account-sourced, covered cases)
The call ranking adds little	per 1,000 decisions: live model +6.2 [+1.1 to +10.5], calling every call decision +11.7 [+4.5 to +18.1], a depth-3 policy tree +13.4 [+9.3 to +17.3]
Kept's choice vs what the bank did	0 per 1,000 [-7 to +8] on a clean outcome; +42 [31–52] on the old label, which counts link payments seen only on the contact row
The bank's recommendation field is look-ahead	it is the last recommendation as of the snapshot, so it encodes how each case ended; not a baseline
The gate's noise understates seed spread	call Qini +0.0023 to +0.0063 over four seeds of the same recipe
Timing is the clear win	evening calls reach the right party +20.9 pts more often [+20.1 to +21.8], +24.0 pts within the same customer; booking to the best hour at today's team volume is predicted to lift right-party contact from 43.4% to 45.1% (+1.7 pts [+1.5 to +1.9]) against the hours the bank called at, moving 84% of calls (best_hour.md; the study's simpler model said +7.4 pts)

Source: docs/reports/uplift_improvement.md, best_hour.md

APPENDIX

Drift and fairness, in detail

Largest population shifts (PSI, live vs training)



Above 0.25 is drift. The promise-kept rate fell from about 63% to 35% in 2026, so cohort rates moved most; drift on a model input pauses promotion.

Fairness flags (four-fifths rule, ratio under 0.80)

gender code, call: 0.72; marital status, call: 0.73; marital status, hardship review: 0.76.

Each is reviewed: small groups, or explained by recorded hardship. Protected attributes are never model inputs, and scrambling them changes 0 of 96,000 cards.

Source: data/exports/reports/drift_data.json, fairness_data.json

APPENDIX

Backend: one machine to 40x

Median, same machine	DuckDB	Polars	PostgreSQL	PySpark
roll_rate 1x	0.9 s	0.5 s	10.4 s	5.3 s
roll_rate 10x	9.8 s	out of memory	skipped (disk)	38.9 s
silver build 1x	11.4 s	2.4 s	9.5 s	10.7 s
silver build 10x	94.3 s	out of memory	skipped (disk)	96.2 s

8 GB memory cap; DuckDB limited to 4 GB and spilling to disk. DuckDB reached 40x on `roll_rate` (61.7 s) and extrapolates to about 42x before the 24 GB limit.

Nightly batch, not hourly

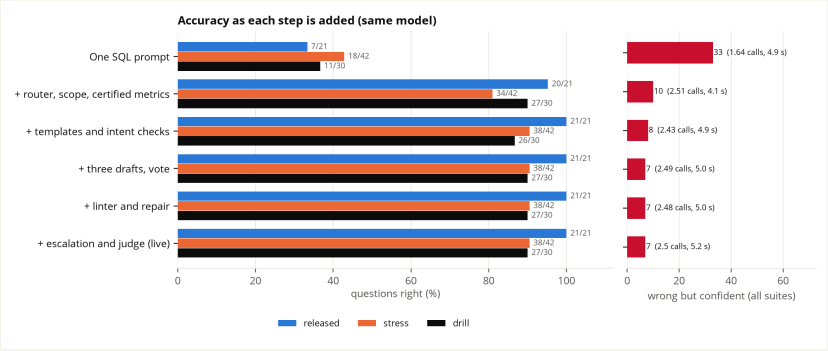
Re-scoring every hour changes 3.1% of cards by the end of a day (2,839 of 90,998); 2.7% of cards change with no new note, which the live-note path does not cover.

One feature definition

A rewrite with the wrong clock differs on 1,325 of 56,000 cells, a wrong time zone on 808; the control on 0. `kept parity` stops both; the clock bug would change 5,503 cards.

APPENDIX

The Ask: the router and certified metrics carry the gain



Same model, 93 graded questions. Router and certified metrics: 38.7% to 87.1%, wrong-but-confident answers 33 to 10.

Templates add 4; drafts, vote, linter and judge add 1.

Source: docs/reports/decisions.md B5; data/exports/reports/decision_B5.json

APPENDIX

Three model choices, measured

Choice	Option	Score	Also
Reading hardship in notes (B7): F1; hours for 760,594 notes	Keyword rules	0.63	recall 0.75; 0.003 h
	SetFit (GPU)	0.87	recall 0.98; 0.018 h
	LLM	0.88	recall 1.00; 701 h (extrapolated)
Ranking contacts (B10), clean outcome: Qini in noise units; extra payers per 1,000 in the top 30% (old label)	Response model	6.6	64 per 1,000 (108)
	T-learner (live)	2.8	87 per 1,000 (142); ahead at the top, behind on Qini
	S-learner	13.5	122 per 1,000 (174); open: pilot decides
	Causal forest	9.3	112 per 1,000 (172)
Promise model (B9): AUC	Six features	0.625	easy to explain
	Wide model	0.697	the bank's score: 0.702

Source: docs/reports/decisions.md B7, B9, B10

APPENDIX

Which local model

Model	GPU MiB	Ask right, of 93	Confident wrong	Notes F1	Brief, s
Qwen3.8-27B (live)	14,733	86	7	0.88	4.79
Qwen3-14B	12,802	74	13	0.905	2.92
Qwen3-8B	8,546	75	17	0.87	1.54
Gemma 3 12B	10,520	71	16	0.739	not run

Only one model fits the 16,303 MiB card, so it serves all three jobs. Gemma 3 12B was stopped after it failed the notes hardship gate. Ollama (same Qwen3-14B file) cut a 9,743-token prompt to 4,096 by default; with the context matched it runs at 78.3 tokens a second against 79.4 on llama.cpp. Live red team: 41 of 43 attacks refused; the other 2 answered only the legitimate part.

Source: docs/reports/decisions.md B1, B3; data/exports/reports/decision_B1.json, decision_B3.json